Day 040 | Unit 03 | Teacher plan

Integers and Rational Numbers

Interpret Points on Axes and in Quadrants

Standards: 6.NS.C.6b, 6.NS.C.6c

Learning target: I can plot ordered pairs in all four quadrants.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Start at the origin. Move horizontally by x, then vertically by y. Quadrant signs are I (+,+), II (-,+), III (-,-), IV (+,-). A point on an axis belongs to no quadrant.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 1

Day 040 | Unit 03 | Lesson

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01 / Learn

Start at the origin. Move horizontally by x, then vertically by y. Quadrant signs are I (+,+), II (-,+), III (-,-), IV (+,-). A point on an axis belongs to no quadrant.

Worked example 1

Plot P(6, -4) on a coordinate plane and name its quadrant.

1. Start at the origin.

2. Move horizontally by x, then vertically by y.

3. Move 6 right, then 4 down.

4. State the result in the form requested: Quadrant IV.

Answer: Quadrant IV

Worked example 2

Plot P(-5, 4), Q(5, -4), and R(0, -4). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.

1. Start at the origin.

2. P has signs (-,+), so II.

3. Q has (+,-), so IV.

4. R has x = 0 and is on an axis.

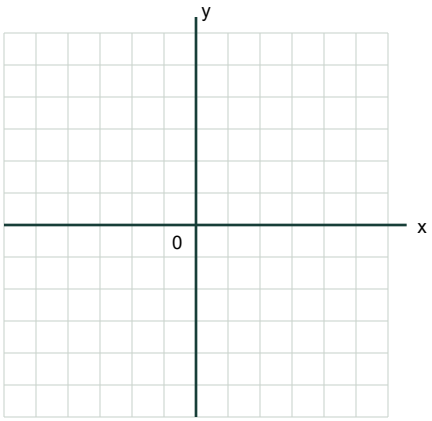
Answer: P: II; Q: IV; R is on the y-axis.

Day 040 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

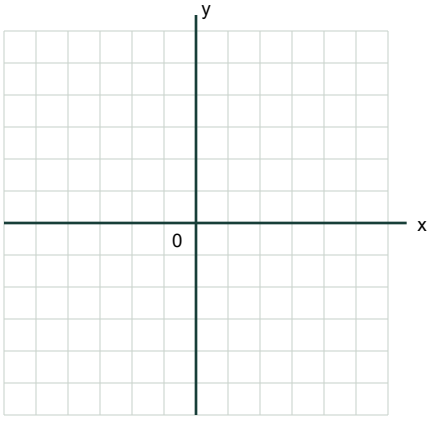
Plot P(-7, -5) on a coordinate plane and name its quadrant.



Label the scale and units.

02 / Constructed response

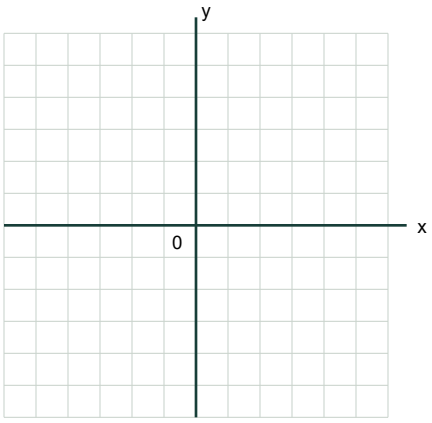
Plot P(5, -3) on a coordinate plane and name its quadrant.



Label the scale and units.

03 / Constructed response

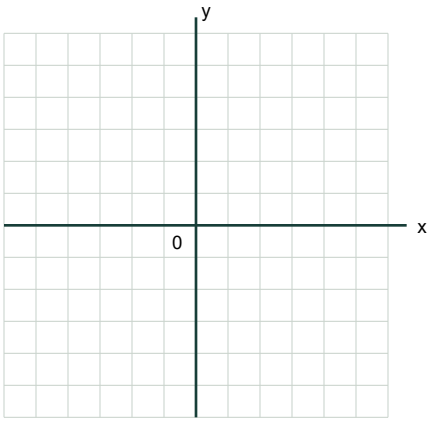
Plot P(-8, 2) on a coordinate plane and name its quadrant.



Label the scale and units.

04 / Constructed response

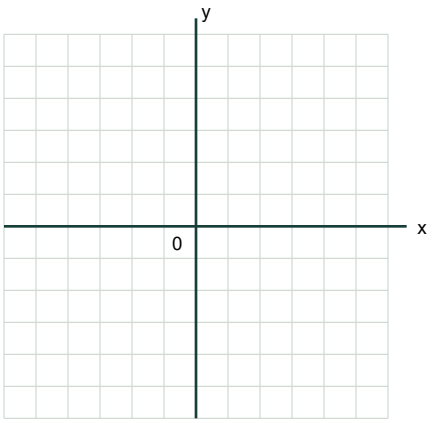
Plot P(5, -6) on a coordinate plane and name its quadrant.



Label the scale and units.

05 / Constructed response

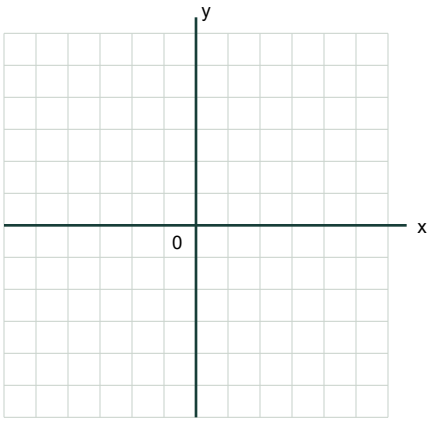
Plot P(-8, -4) on a coordinate plane and name its quadrant.



Label the scale and units.

06 / Constructed response

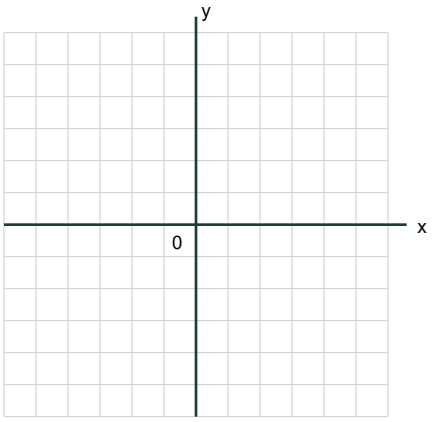
Plot P(3, 4) on a coordinate plane and name its quadrant.



Label the scale and units.

07 / Constructed response

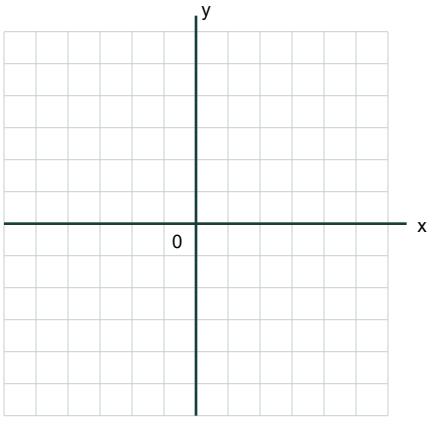
Plot P(-3/2, -1) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



Label the scale and units.

08 / Coordinate plotting and explanation

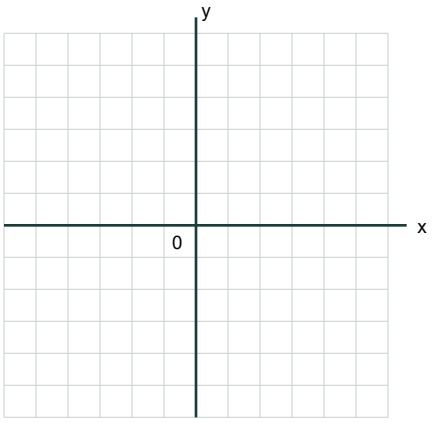
Plot P(-4, 5), Q(4, -5), and R(0, -5). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

09 / Coordinate plotting and explanation

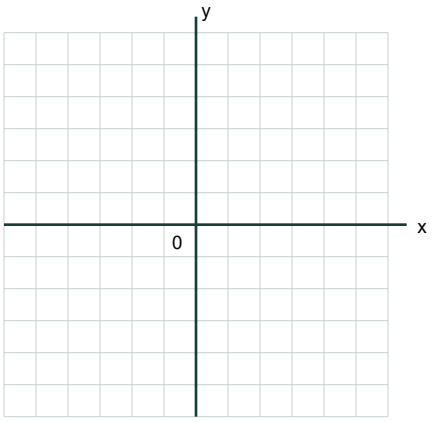
Plot P(-4, 6), Q(4, -6), and R(0, -6). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

10 / Constructed response

Plot P(1, -3/2) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



Label the scale and units.

Day 040 | Unit 03 | Answer key

Integers and Rational Numbers

Interpret Points on Axes and in Quadrants

Standards: 6.NS.C.6b, 6.NS.C.6c

Learning target: I can plot ordered pairs in all four quadrants.

1. Quadrant III

Move 7 left, then 5 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

2. Quadrant IV

Move 5 right, then 3 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

3. Quadrant II

Move 8 left, then 2 up.

6.NS.C.6b, 6.NS.C.6c | Constructed response

4. Quadrant IV

Move 5 right, then 6 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

5. Quadrant III

Move 8 left, then 4 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

6. Quadrant I

Move 3 right, then 4 up.

6.NS.C.6b, 6.NS.C.6c | Constructed response

7. Quadrant III; Q(0, -1)

Move 3/2 left, then 1 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

8. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

9. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

10. Quadrant IV; Q(0, -3/2)

Move 1 right, then 3/2 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response